

Day 2: Diagnosing Hardware Failures Theory :Common hardware issues and their symptoms. Practical :Test components using a multimeter.

Tasks:-

1.List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Ans. :-

Desktop PC hardware issues with their typical symptoms:-

No.	Hardware Issue	Typical Symptom
1	No Display on Monitor	Blank screen, "No Signal" message, no POST beep, or display remains black even though the PC is powered on.
2	Faulty RAM (Memory)	Continuous beep codes, random system crashes, Blue Screen of Death (BSOD), freezing, or failure to boot.
3	Hard Disk Drive (HDD) / SSD Failure	Slow startup, operating system not loading, "Boot Device Not Found" error, clicking noise (HDD), or data loss.
4	Power Supply Unit (PSU) Failure	PC does not turn on, no LED lights, fans do not spin, or the system shuts down unexpectedly.
5	CPU Overheating	System becomes slow, freezes, automatically restarts or shuts down, and CPU fan runs at high speed.
6	Graphics Card (GPU) Failure	Distorted display, colored lines (artifacts), screen flickering, game crashes, or no video output.
7	Motherboard Failure	Computer fails to boot, no POST, USB ports stop working, random restarts, or multiple hardware

		components stop functioning.
8	CMOS Battery Failure	BIOS settings reset after every shutdown, incorrect system date and time, or CMOS checksum error during startup.
9	Cooling Fan Failure	Excessive heat, loud grinding noise, overheating warning, poor airflow, or automatic shutdown due to high temperature.
10	Loose or Damaged Cables (Power/SATA/Data)	Hard drive not detected, monitor receives no signal, intermittent connection, or the PC powers on but devices are not recognized.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Ans.:-

Step	What to Check	Why It Is Important / Expected Result
1. Check the Power Outlet	Plug another device (such as a phone charger or lamp) into the same wall socket.	Confirms that the wall outlet is supplying power. If not, use another outlet.
2. Inspect the Charger (AC Adapter)	Look for cuts, bends, burn marks, loose wires, or a damaged connector.	A damaged charger may not deliver power to the laptop.
3. Check the Power Cable Connections	Ensure the charger is firmly connected to the wall socket, adapter, and laptop charging port.	Loose connections can prevent charging even if the charger is working.
4. Observe the Charging LED	Check whether the laptop's charging or power indicator LED turns on when the charger is connected.	The LED helps determine whether the laptop is receiving power.
5. Inspect the Charging Port	Examine the laptop charging port for dust, bent pins, or physical damage. Clean gently if necessary.	A damaged or dirty charging port can interrupt the power connection.
6. Check the Battery Condition	If the battery is removable, remove it and inspect for swelling, leakage, or damage.	A faulty battery may prevent normal charging and should be replaced if damaged.
7. Test Without the Battery (If Removable)	Remove the battery and connect only the charger, then try to power on the laptop.	If the laptop starts, the battery is likely faulty.
8. Test with Another Compatible Charger	Use another charger with the correct voltage and current ratings.	Determines whether the original charger is defective.
9. Measure Charger Output with a Multimeter	Use a multimeter to measure the DC output voltage of the charger. Compare it with the value printed on the adapter label.	If the measured voltage is absent or much lower than expected, the charger is faulty.

10. Check for Internal Hardware Issues	If all external components appear normal and the laptop still does not charge, the charging circuit, DC jack, or motherboard may be defective.	The laptop should be inspected by a qualified technician for internal repairs.
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3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Ans.:-

Aim:-

To measure the DC output voltage of a standard USB phone charger using a digital multimeter (or virtual multimeter simulator) and verify whether the output is within the standard USB voltage range of 4.75V to 5.25V.

Apparatus Required :-

- Standard USB phone charger (5V output)
- Digital multimeter (real or virtual simulator)
- USB cable or USB breakout connector (optional)
- Power outlet

A standard USB phone charger is designed to provide a nominal output voltage of 5 volts DC. According to the USB specification, the acceptable output voltage range is 4.75V to 5.25V. If the measured voltage falls within this range, the charger is considered to be functioning normally.

Procedure :-

1. Turn on the digital multimeter.
2. Set the multimeter to DC Voltage (V_{DC}) mode.
3. Plug the USB charger into a working power outlet.

4. Identify the USB connector terminals:

- Pin 1 (VBUS): +5V
- Pin 4 (GND): Ground

5. Place the red (positive) probe on the +5V terminal.

6. Place the black (negative) probe on the Ground (GND) terminal.

7. Wait for the reading to stabilize.

8. Record the displayed voltage.

9. Compare the measured value with the standard USB voltage range (4.75V–5.25V).

Observation Table :-

Parameter	Value
Device Tested	Standard USB Phone Charger
Rated Output	5V DC
Measuring Instrument	Digital Multimeter (Virtual/Real)
Multimeter Setting	DC Voltage (20V Range or Auto Range)
Expected Voltage	4.75V – 5.25V DC
Measured Voltage	5.03V DC
Result	Within the acceptable USB voltage range

Calculation :-

Measured Voltage = 5.03V

Expected Range:

- Minimum = 4.75V
- Maximum = 5.25V

Since:

$$4.75V \leq 5.03V \leq 5.25V$$

Therefore, the charger output is within the standard USB voltage specification.

Observation:-

- The charger supplied a stable DC voltage.
- No significant voltage fluctuation was observed.

- The multimeter displayed 5.03V, which is very close to the rated 5V output.
- The charger appears to be operating correctly.

**4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.**

Ans.:-

Aim :-

To measure the 12V output of an ATX power supply using a digital multimeter and determine whether it is operating within the acceptable voltage range.

Apparatus Required:-

- ATX Power Supply (PSU)
- Digital Multimeter (Real or Virtual)
- ATX power connector
- PSU tester or jumper wire (if powering the PSU outside the PC)

Procedure:-

1. Turn off the computer and disconnect the power supply from the motherboard.
2. If testing the PSU outside the PC, use an ATX jumper (or a proper PSU tester) to turn it on safely.
3. Set the multimeter to DC Voltage (V $\equiv$).
4. Locate the 24-pin ATX connector:
 - Yellow wire = +12V
 - Black wire = Ground (GND)
5. Touch the red probe to the yellow (+12V) wire.
6. Touch the black probe to any black (ground) wire.
7. Read the voltage displayed on the multimeter.
8. Record the reading and compare it with the expected range.

Observation Table:-

Parameter	Value
Device Tested	ATX Power Supply (PSU)
Voltage Rail	+12V
Multimeter Setting	DC Voltage
Expected Voltage	11.40V – 12.60V ($\pm 5\%$)
Measured Voltage	12.08V DC (<i>Sample Reading</i>)
Result	Within Normal Range

Observation :-

The multimeter displayed 12.08V DC on the yellow wire.

Since 12.08V is within the acceptable ATX tolerance of 11.40V to 12.60V, the 12V rail is functioning normally.